

Chi Coupling Selection Rules in TmFeO₃ Γ_2 Phase

Ground-State Lambda Generation and qAFM Cross-Peak Detectability
under $H \parallel c$ (2DCS Geometry)

June 2026

Contents

1	Overview and main result	1
2	Sigma vectors and Bertaut orthogonality	2
3	Fe and Tm local frames	2
3.1	Fe sublattice frames	2
3.2	Tm sublattice frames	3
3.3	Bond connectivity: each Tm sees all four Fe	3
4	Ground-state analysis: which channels activate λ^2	3
4.1	Gamma ₂ Fe spin background	3
4.2	Effective field on λ^2 from each chi channel	3
4.3	Chi5 and chi7: exact cancellation by the inversion sign	4
5	The $H \parallel c$ detector coupling	6
6	Equation-of-motion analysis for the qAFM kick	6
6.1	SU(3) equation of motion	6
6.2	The qAFM kick	7
6.3	Time evolution of $H_\chi[\lambda^2]$ after the kick	7
6.4	Complete channel survey	7
7	What would allow the cross-peak?	8
8	Numerical verification	9
9	Summary diagram	9
A	Bond-by-bond verification: each Tm sees all four Fe	10
B	Sigma vector inner product table	11

1 Overview and main result

We analyse all nine $\chi_{2\alpha}$, $\chi_{5\alpha}$, $\chi_{7\alpha}$ coupling channels ($\alpha \in \{x, y, z\}$) of the mixed Fe–Tm Hamiltonian in TmFeO₃ in its Γ_2 (G_z , F_x) magnetic phase. The central question is whether the

cross-peak at excitation frequency ω_{qAFM} and detection frequency $\omega_{E_{12}}$ (the Tm $|E_1\rangle \rightarrow |E_2\rangle$ transition) can be nonzero in a $H \parallel c$ 2DCS experiment.

Result: The cross-peak ($\omega_{\text{qAFM}}, \omega_{E_{12}}$) is *doubly forbidden* for every chi channel by the following two independent Bertaut-vector orthogonality constraints:

1. **Source failure:** in the Γ_2 phase no chi channel produces σ_C -pattern λ^2 . Every channel produces either zero or σ_F -pattern λ^2 .
2. **Detector failure:** $H \parallel c$ couples to $M_z^{\text{Tm}} = c_2 (\sigma_C \cdot \lambda_{\text{vec}}^2)$. Since $\sigma_C \cdot \sigma_F = 0$, the signal vanishes even when λ^2 is large.

The proof rests on two exact algebraic identities of the Bertaut sigma vectors (Sec. 2) together with the bond connectivity and local-frame conventions of `unitcell_builders.cpp` (Sec. 3). Numerical verification is provided by the diagnostic executable `diag_tmfeo3_qafm_e12_xpeak` (Sec. 8).

2 Sigma vectors and Bertaut orthogonality

The four Pbnm sublattice-staggering patterns are defined as four-component vectors in the order ($\mu = 0, 1, 2, 3$):

$$\sigma_F = (+, +, +, +), \quad \sigma_G = (+, -, +, -), \quad \sigma_A = (+, +, -, -), \quad \sigma_C = (+, -, -, +). \quad (1)$$

They form an orthogonal set under the sublattice inner product $\sigma_P \cdot \sigma_Q = \sum_{\mu=0}^3 \sigma_P[\mu] \sigma_Q[\mu]$:

$$\sigma_P \cdot \sigma_Q = 4 \delta_{PQ}, \quad P, Q \in \{F, G, A, C\}. \quad (2)$$

The component-wise product rule (Hadamard product) closes within the set:

$$\sigma_G \circ \sigma_G = \sigma_F, \quad \sigma_A \circ \sigma_G = \sigma_C, \quad \sigma_C \circ \sigma_G = \sigma_A, \quad \sigma_A \circ \sigma_A = \sigma_F, \quad \sigma_C \circ \sigma_C = \sigma_F, \quad \dots \quad (3)$$

3 Fe and Tm local frames

3.1 Fe sublattice frames

The four Fe sublattices sit at Wyckoff 4b positions. Their local frames (stored as `R_fe_frames` in `unitcell_builders.cpp`) are diagonal:

$$R_{\text{Fe}}^{(k)} = \text{diag}(\eta_k^x, \eta_k^y, \eta_k^z), \quad k = 0, 1, 2, 3. \quad (4)$$

The diagonal entries and their sublattice patterns are:

	Fe ₀ (E)	Fe ₁ (C_{2x})	Fe ₂ (C_{2z})	Fe ₃ (C_{2y})
η^x	+1	+1	-1	-1
η^y	+1	-1	-1	+1
η^z	+1	-1	+1	-1
pattern	$\eta_{\text{Fe}}^x = \sigma_A$ $\eta_{\text{Fe}}^y = \sigma_C$ $\eta_{\text{Fe}}^z = \sigma_G$			

The Fe spin is stored in the global lab frame, so the local component along axis α is $S_{\text{loc},k}^\alpha = R_{\alpha\alpha}^{(k)} S_k^\alpha = \eta_k^\alpha S_k^\alpha$.

3.2 Tm sublattice frames

The four Tm sublattices sit at Wyckoff 4c positions. Their local frames (from `tmfeo3_tm_local_frames_xyz()`) are also diagonal:

	Tm ₀	Tm ₁	Tm ₂	Tm ₃
η^x	+1	+1	-1	-1
η^y	+1	-1	+1	-1
η^z	+1	-1	-1	+1
pattern	$\eta_{\text{Tm}}^x = \sigma_A$ $\eta_{\text{Tm}}^y = \sigma_G$ $\eta_{\text{Tm}}^z = \sigma_C$			

The physical Tm magnetic moment along the global z -axis is

$$J_{j,\text{glob}}^z = \eta_{\text{Tm},j}^z J_{j,\text{loc}}^z = \sigma_C[j] c_2 \lambda_j^2, \quad (5)$$

where $c_2 = \langle E_1 | J^z | E_2 \rangle_{\text{loc}}$ is the local Tm CEF matrix element, and λ^2 is the imaginary part of the $|E_1\rangle$ - $|E_2\rangle$ coherence (second Gell-Mann component).

3.3 Bond connectivity: each Tm sees all four Fe

From `fe_tm_w_bond_pairs()` there are 4 orbits $\times$ 2 bonds (even/odd) = 16 bond pairs in the unit cell. A key property is:

Each Tm sublattice j has exactly one bond per orbit, connecting it to each of the four Fe sublattices. (6)

Consequently, summing the chi coupling over all bonds to Tm _{j} is equivalent to the inner product over all Fe sublattices:

$$H_{\chi,j}^\alpha[\lambda^2] = \chi_{2\alpha} \sum_{k=0}^3 \eta_{\text{Fe},k}^\alpha S_{\text{Fe},k}^\alpha. \quad (7)$$

Figure 1 shows the 3D crystal structure with all 16 even bonds coloured by orbit. Figure 2 shows the Fe local frame axes (arrows) at each sublattice position.

4 Ground-state analysis: which channels activate λ^2

4.1 Gamma₂ Fe spin background

In the Γ_2 (G_z , F_x) magnetic phase:

$$S_{\text{Fe},k}^z = G_z \sigma_G[k], \quad S_{\text{Fe},k}^x = F_x \sigma_F[k], \quad S_{\text{Fe},k}^y = 0, \quad (8)$$

where $G_z \approx 2.5 \mu_B$ and $F_x \approx 0.014 \mu_B$ ($F_x/G_z \approx 0.006$, set by the DM ratio D_1/H_E).

4.2 Effective field on λ^2 from each chi channel

Substituting (8) into (7):

$$H[\lambda^2]_j^{\chi_{2z}} = \chi_{2z} \sum_k \sigma_G[k] \cdot G_z \sigma_G[k] = \chi_{2z} G_z \sum_k \sigma_G[k]^2 = 4\chi_{2z} G_z. \quad (9)$$

$$H[\lambda^2]_j^{\chi_{2x}} = \chi_{2x} \sum_k \sigma_A[k] \cdot F_x \sigma_F[k] = \chi_{2x} F_x (\sigma_A \cdot \sigma_F) = 0. \quad (10)$$

$$H[\lambda^2]_j^{\chi_{2y}} = \chi_{2y} \sum_k \sigma_C[k] \cdot 0 = 0. \quad (11)$$

Key observations:

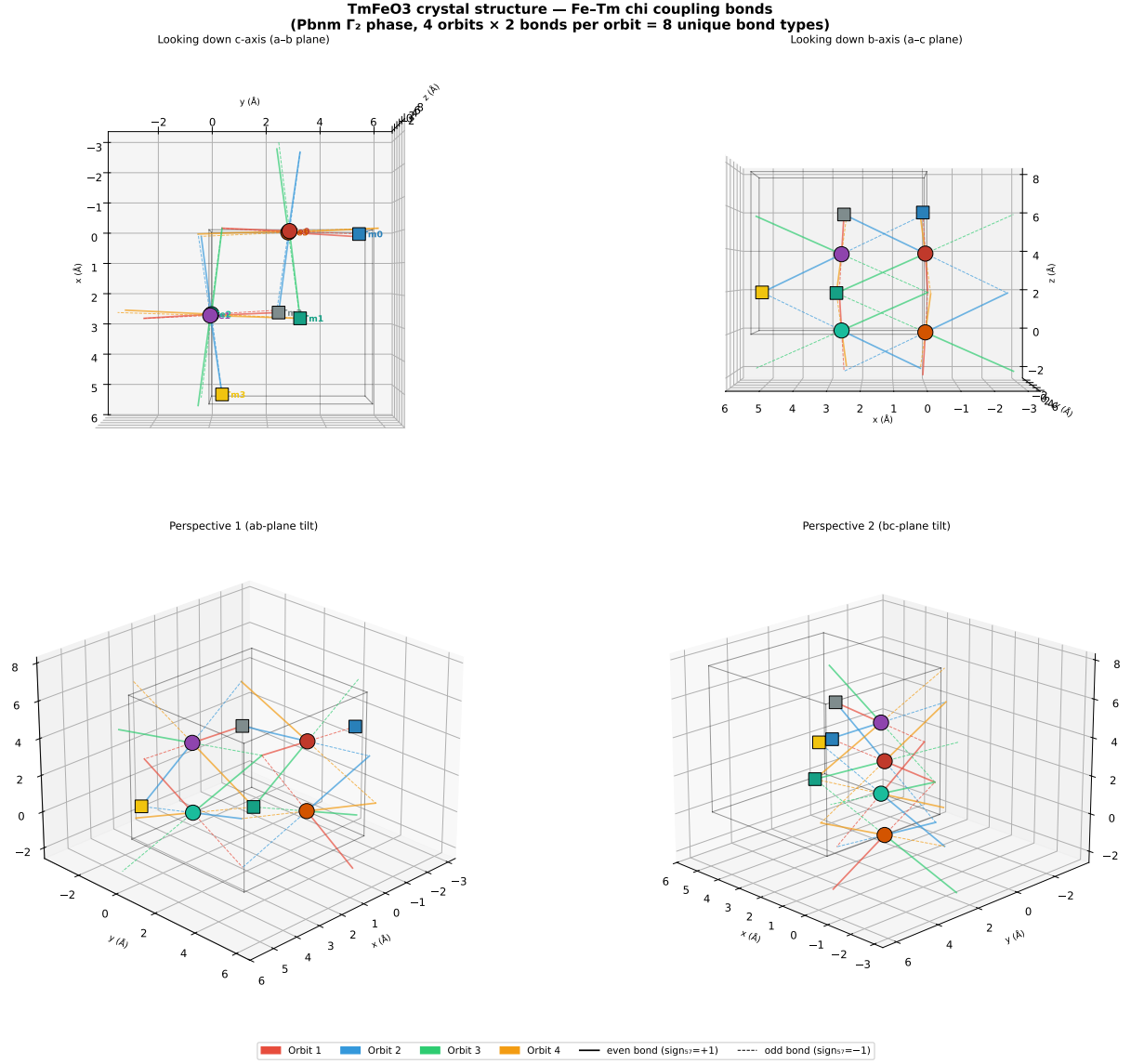


Figure 1: TmFeO₃ crystal structure with Fe–Tm chi coupling bonds. Circles: Fe sublattices (red/purple/teal/orange = sub 0/1/2/3). Squares: Tm sublattices (blues/greys/yellow). Solid lines: even bonds (sign₅₇ = +1); dashed: odd bonds (sign₅₇ = -1). Colours: orbit 1 (red), 2 (blue), 3 (green), 4 (orange). Each Tm sublattice has exactly one bond per orbit, connecting it to all four Fe sublattices.

- Equation (9): $\sigma_G[k]^2 = 1$ for each k , so the sum is +4 at *every* Tm sublattice identically. The resulting λ_j^2 is **uniform** (σ_F -pattern). Numerical value: $\lambda^2 \approx -0.997 \sigma_F$ at equilibrium (from `diag_tmfeo3_qafm_e12_xpeak`).
- Equation (10): $\sigma_A \cdot \sigma_F = \sum_k (+, +, -, -)(+, +, +, +) = 0$. The small F_x canting is invisible to χ_{2x} .
- Equation (11): $S_{\text{Fe}}^y = 0$ in Γ_2 , so χ_{2y} is completely inert.

4.3 Chi5 and chi7: exact cancellation by the inversion sign

For $\chi_{5\alpha}$ (coupling to λ^5), the bond tensor acquires a factor sign₅₇ = ± 1 : the even bond of each orbit pair carries +1, and the odd (inversion-related) bond carries -1. Crucially, within each

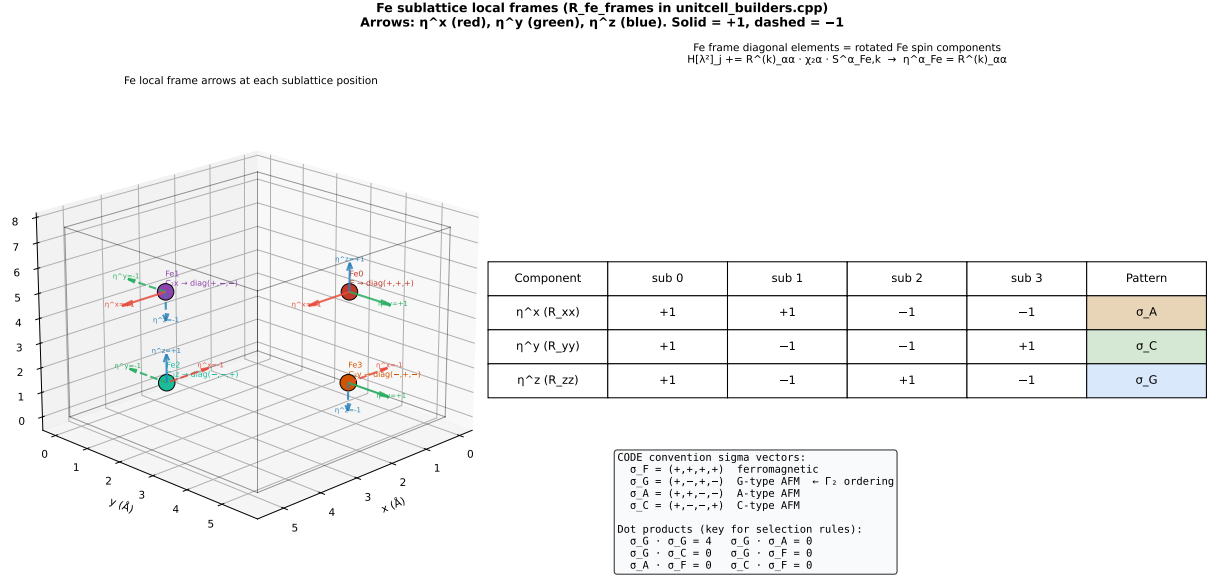


Figure 2: *Left*: Fe local frame arrows at each sublattice position. Red = $\hat{x}$, green = $\hat{y}$, blue = $\hat{z}$. Solid arrow: $\eta = +1$; dashed: $\eta = -1$. *Right*: Summary table of Fe frame diagonal elements and their Bertaut patterns. The right-hand panel also lists the Bertaut inner products most relevant for the selection rules.

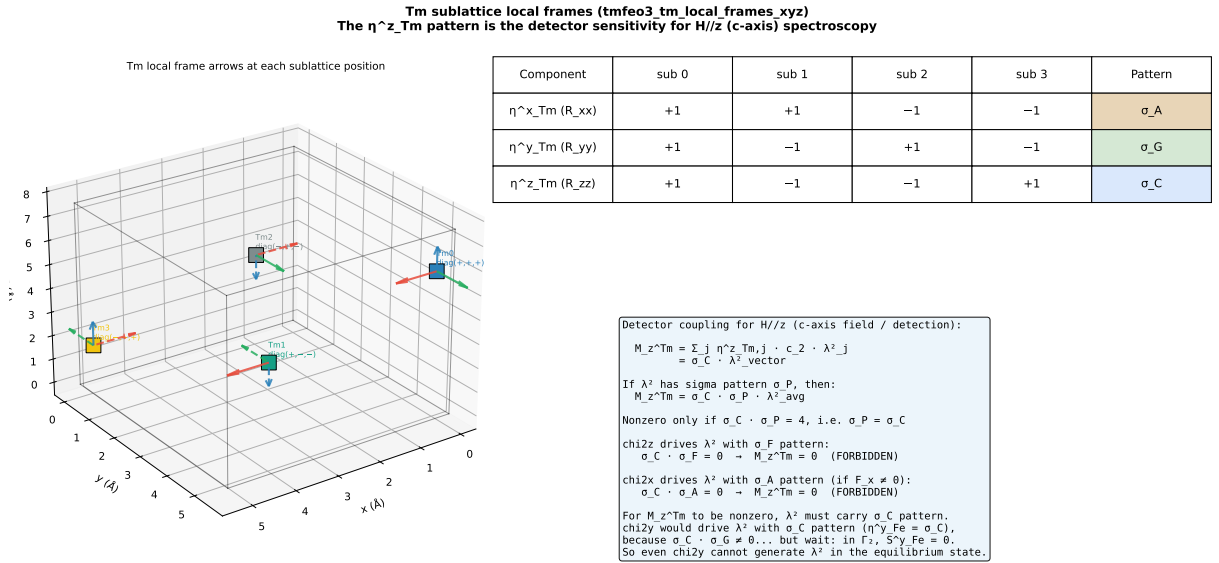


Figure 3: *Left*: Tm local frame arrows. *Right*: Tm frame patterns and the detector coupling analysis. $\hat{\eta}^z_{Tm} = \sigma_C = (+, -, -, +)$, so $H \parallel c$ detects $M_z^{Tm} = c_2 (\sigma_C \cdot \lambda^2_{vec})$. For this to be nonzero, λ^2 must carry σ_C pattern.

orbit pair, *both bonds connect the same Fe sublattice to (different) Tm sites*. Therefore the net contribution per Tm site is:

$$H[\lambda^5]_j \propto (+1) \eta_{Fe,k}^\alpha S_k^\alpha + (-1) \eta_{Fe,k}^\alpha S_k^\alpha = 0. \quad (12)$$

The cancellation is exact for all Fe spin configurations (χ_5 and χ_7 are dead at $q = 0$ for *any* magnetic order).

5 The $H \parallel c$ detector coupling

The Tm contribution to the z -magnetisation observable is (from Eq. (5)):

$$M_z^{\text{Tm}} = g_J \mu_B \sum_{j=0}^3 \sigma_C[j] c_2 \lambda_j^2 = g_J \mu_B c_2 (\sigma_C \cdot \lambda_{\text{vec}}^2). \quad (13)$$

From the ground-state result (Sec. 4), $\lambda_j^2 \approx \bar{\lambda}^2 \sigma_F[j]$, so:

$$M_z^{\text{Tm}} = g_J \mu_B c_2 \bar{\lambda}^2 (\sigma_C \cdot \sigma_F) = g_J \mu_B c_2 \bar{\lambda}^2 \cdot 0 = 0, \quad (14)$$

using $\sigma_C \cdot \sigma_F = (+, -, -, +) \cdot (+, +, +, +) = 1 - 1 - 1 + 1 = 0$.

This is the first selection rule: **even though χ_{2z} generates large $\lambda^2 \approx \pm 1$, the signal M_z^{Tm} is identically zero because the σ_F pattern of λ^2 is orthogonal to the σ_C detector pattern.**

Chi coupling channel activity in TmFeO3 Gamma_2 phase
Left: ground-state lambda generation | Right: H/z detectability (M_z^{Tm})

Ground state: $H \parallel c$ pattern generated by each channel $M_z^{\text{Tm}} = \eta^z \lambda^2 \cdot \text{Tm} \cdot \lambda^2 = \sigma_C \cdot \lambda^2$ (H/z detector coupling)

Channel	$\eta^z \alpha_{\text{Fe}}$	Fe spin pattern	η^z S product	λ^2 sigma pattern	Active?	Channel	λ^2 sigma pattern	$\sigma_C \cdot \lambda^2$ sigma	M_z^{Tm}	Detectable by H/z?
chi2z	$\alpha_G = (+ - -)$	$G_z \alpha_G$	$\alpha_G \alpha_F = \alpha_F$	α_F (uniform)	YES ★	chi2z	$\alpha_F = (++++)$	$\sigma_C \cdot \alpha_F = 0$	$= 0$	NO ✗
chi2x	$\alpha_A = (++ -)$	$F_x \alpha_A$	$\alpha_A \alpha_F = 0$	none ($F_x = 0$)	no	chi2x	α_A (tiny)	$\sigma_C \cdot \alpha_A = 0$	$= 0$	NO ✗
chi2y	$\alpha_C = (+ - +)$	$S^y = 0$	—	none	no	chi2y	dead	—	0	NO ✗
chi5z	$\alpha_G = (+ - -)$	(any)	cancels	none	no ✗	chi5*	dead	—	0	NO ✗
chi5x	$\alpha_A = (++ -)$	(any)	cancels	none	no ✗	chi7*	dead	—	0	NO ✗
chi5y	$\alpha_C = (+ - +)$	(any)	cancels	none	no ✗	—	need α_C	$\sigma_C \cdot \alpha_C = 4$	$\neq 0$	YES
chi7z	$\alpha_G = (+ - -)$	(any)	cancels	none	no ✗					
chi7x	$\alpha_A = (++ -)$	(any)	cancels	none	no ✗					
chi7y	$\alpha_C = (+ - +)$	(any)	cancels	none	no ✗					

★ chi5/chi7 cancel because even/odd bonds share the same Fe substitution within each orbit pair sign. $S^y = (+1 \times -1) = 0$ identically.

chi2y inactive because $S^y = 0$ in Γ_1 by symmetry.

chi2x inactive because $\alpha_A \alpha_F = 0$ (small F_x canting is α_F , not α_A).

H/z couples to $M_z^{\text{Tm}} = \sum_j \eta^z \lambda_j^2 \cdot \text{Tm} \cdot \lambda_j^2$ (local)

$[j^z, \text{Tm}]_{\text{local}} = c_2 \cdot \lambda_j^2$ (leading-order matrix element)

$\eta^z \lambda^2 \cdot \text{Tm} = \sigma_C = (+, -, -, +)$ [from Tm frame diagonals R_{zz}]

For any chi channel, λ^2 must carry σ_C pattern to be detected by H/z.

No chi2 channel produces σ_C lambda² from Gamma_2 background:

chi2z $\rightarrow \alpha_F$, chi2x $\rightarrow 0$, chi2y $\rightarrow$ dead.

CONCLUSION: The (omega, qAFM, omega_E12) cross-peak is FORBIDDEN under H/z for ALL chi channels in the Gamma_2 phase.

Figure 4: Summary table. *Left*: ground-state lambda generation for each chi channel in the Γ_2 background. *Right*: detectability by $H \parallel c$ ($M_z^{\text{Tm}} = \sigma_C \cdot \lambda^2$). Only σ_C -pattern λ^2 would be detectable; no channel generates it.

6 Equation-of-motion analysis for the qAFM kick

6.1 SU(3) equation of motion

The Tm^{3+} non-Kramers ion is modelled as a three-level system. Its density matrix is expanded in the Gell-Mann basis $\{F_a = \lambda^a/2\}$:

$$\rho_j = \frac{1}{3} \mathbf{1} + \frac{1}{2} \sum_{a=1}^8 \lambda_j^a F_a. \quad (15)$$

The Liouville equation $\dot{\rho} = -i[H_{\text{eff}}, \rho]/\hbar$ projected onto the Gell-Mann basis gives:

$$\dot{\lambda}_j^a = \frac{1}{\hbar} \sum_{b,c} f^{abc} H_{\text{eff},j}^b \lambda_j^c, \quad (16)$$

where f^{abc} are the totally antisymmetric SU(3) structure constants and

$$H_{\text{eff},j}^b = H_{\text{CEF}}^b + H_{\chi,j}^b + H_{\text{Jtm},j}^b + H_{\text{field},j}^b. \quad (17)$$

For the chi coupling alone (setting all other terms to zero):

$$H_{\chi,j}^{\lambda^2} = \sum_{k \in \text{bonds}(j)} \eta_{\text{Fe},k}^z \chi_{2z} S_{\text{Fe},k}^z, \quad (18)$$

and similarly for χ_{2x} , χ_{2y} with the appropriate η_{Fe}^α pattern.

The λ^2 component represents the imaginary part of the $|E_1\rangle - |E_2\rangle$ coherence. Its equation of motion couples it to λ^1 (the real part of the same coherence) through the CEF Hamiltonian, forming the E_{12} oscillation.

6.2 The qAFM kick

A qAFM kick rotates all Fe spins by $\delta\theta$ around the y -axis:

$$S_{\text{Fe},k}^x(0^+) = G_z \sin \delta\theta \cdot \sigma_G[k] + F_x \cos \delta\theta \cdot \sigma_F[k], \quad (19)$$

$$S_{\text{Fe},k}^z(0^+) = G_z \cos \delta\theta \cdot \sigma_G[k] - F_x \sin \delta\theta \cdot \sigma_F[k]. \quad (20)$$

For $\delta\theta \ll 1$, the dominant new component is a G_x perturbation:

$$\delta S_{\text{Fe},k}^x \approx G_z \delta\theta \sigma_G[k]. \quad (21)$$

The kick thus introduces a G_x component with pattern σ_G .

6.3 Time evolution of $H_\chi[\lambda^2]$ after the kick

Chi2z. After the kick, $S_{\text{Fe},k}^z(t) = G_z \cos(\omega_{\text{qAFM}}t) \sigma_G[k]$ (qAFM oscillation in G_z , neglecting damping). Equation (18) gives:

$$H_{\chi_{2z},j}[\lambda^2](t) = \chi_{2z} \sum_k \sigma_G[k] \cdot G_z \cos(\omega_{\text{qAFM}}t) \sigma_G[k] = 4\chi_{2z} G_z \cos(\omega_{\text{qAFM}}t). \quad (22)$$

This is *uniform* across all Tm sublattices. The resulting $\lambda_j^2(t)$ inherits the σ_F pattern at all times, so $M_z^{\text{Tm}}(t) = c_2(\sigma_C \cdot \sigma_F) \cdot \lambda^2(t) = 0$ identically.

The kick also creates $G_x \propto \sigma_G$. Does G_x enter χ_{2z} ? No: χ_{2z} couples only to S_{Fe}^z , not S_{Fe}^x .

Chi2x. The G_x component from the kick is in σ_G pattern. The chi2x coupling is:

$$H_{\chi_{2x},j}[\lambda^2](t) = \chi_{2x} \sum_k \sigma_A[k] \cdot G_x \cos(\omega_{\text{qAFM}}t) \sigma_G[k] = \chi_{2x} G_x \cos(\omega_{\text{qAFM}}t) \underbrace{(\sigma_A \cdot \sigma_G)}_{=0} = 0. \quad (23)$$

Chi2x has $\eta_{\text{Fe}}^x = \sigma_A$ and the kick pattern is σ_G : $\sigma_A \cdot \sigma_G = (+, +, -, -) \cdot (+, -, +, -) = 1 - 1 - 1 + 1 = 0$. So chi2x is blind to the qAFM kick.

Chi2y. $S_{\text{Fe}}^y = 0$ is unaffected by the y -axis rotation (to linear order), so chi2y produces nothing.

6.4 Complete channel survey

For the cross-peak (ω_{qAFM} , $\omega_{E_{12}}$) to be nonzero under $H \parallel c$, we need:

$$M_z^{\text{Tm}}(t) \sim e^{i\omega_{\text{qAFM}}t}, \quad \text{i.e. } \sigma_C \cdot \lambda_{\text{vec}}^2(t) \neq 0. \quad (24)$$

This requires λ_j^2 to carry σ_C pattern. For a chi channel with $\eta_{\text{Fe}}^z = \sigma_G$ (chi2z) to produce σ_C -pattern λ^2 , we would need the Fe spin component to carry $\sigma_G \circ \sigma_C^{-1} = \sigma_G \circ \sigma_C = \sigma_A$ pattern (using $\sigma_G \circ \sigma_C = \sigma_A$ from Eq. (3)). But $A_z = 0$ in Γ_2 .

Table 1 compiles the complete failure analysis for all channels contributing to λ^2 dynamics after the qAFM kick.

Table 1: Chi channel activity for the $(\omega_{\text{qAFM}}, \omega_{E_{12}})$ cross-peak under $H \parallel c$. “Source” = which Fe pattern drives λ^2 . “ $\sigma_C \cdot \lambda^2$ ” = whether the $H \parallel c$ detector sees it.

Channel	η_{Fe}^α	Fe pattern after kick	$\eta \cdot S$ product	λ^2 pattern	$\sigma_C \cdot \lambda^2$
χ_{2z}	σ_G	$G_z \sigma_G$ (and $G_x \sigma_G$)	$\sigma_G \circ \sigma_G = \sigma_F$	σ_F	$\sigma_C \cdot \sigma_F = 0$ FAIL
χ_{2x}	σ_A	$G_x \sigma_G$	$\sigma_A \cdot \sigma_G = 0$	none	0 FAIL
χ_{2y}	σ_C	$S^y = 0$	—	none	0 FAIL
$\chi_{5\alpha}$	any	any	$+1 - 1 = 0$ (sign ₅₇)	none	0 FAIL
$\chi_{7\alpha}$	any	any	$+1 - 1 = 0$ (sign ₅₇)	none	0 FAIL
<i>needed</i>		$S^z \sim \sigma_A$ or $S^y \sim \sigma_F$	gives σ_C	σ_C	$\sigma_C \cdot \sigma_C = 4$ OK

7 What would allow the cross-peak?

The obstacle is a two-layer orthogonality:

$$\underbrace{\sigma_G}_{\text{Fe frame (chi2z)}} \circ \underbrace{\sigma_G}_{\text{Gamma}_2 \text{ order}} = \sigma_F, \quad \underbrace{\sigma_C}_{\text{Tm frame}} \cdot \sigma_F = 0. \quad (25)$$

To make the cross-peak observable one needs to break one of these two links. We list the options below.

Option A: Change the Fe ordering

If $S_{\text{Fe}}^z \sim \sigma_A$ (A-type AFM, Γ_1 phase), then:

$$H[\lambda^2]_j^{\chi_{2z}} = \chi_{2z} G_z (\sigma_G \circ \sigma_A) = \chi_{2z} G_z \sigma_C. \quad (26)$$

Now $\lambda^2 \sim \sigma_C$ and $\sigma_C \cdot \sigma_C = 4 \neq 0$: the cross-peak is allowed in the Γ_1 phase under χ_{2z} .

Alternatively, a G_y ordering (rotating σ_G from z to y) would require χ_{2y} to be active ($\eta_{\text{Fe}}^y = \sigma_C$, and $\sigma_C \circ \sigma_G = \sigma_A$, still not σ_C). In fact, for $H \parallel c$ detection via chi2, the only working combination is:

$$\text{chi2z} + A\text{-type order } (\Gamma_1).$$

Option B: The complete bond connectivity locks in σ_F pattern

Even in the most general case with all chi channels alive simultaneously, the effective field on Tm_j is:

$$H[\lambda^2]_j = \sum_{\alpha \in \{x,y,z\}} \chi_{2\alpha} \underbrace{\sum_{k=0}^3 \eta_{\text{Fe},k}^\alpha S_{\text{Fe},k}^\alpha}_{\text{global sum, no } j \text{ dependence}}. \quad (27)$$

Because *every* Tm sublattice connects to *all four* Fe sublattices (Appendix A), the inner sum over k is the same scalar for every j . The λ^2 field is therefore always uniform (σ_F -pattern), regardless of which $\chi_{2\alpha}$ are nonzero:

$$H[\lambda^2]_j = \underbrace{\chi_{2z} G_z (\sigma_G \cdot \sigma_G)}_{=4\chi_{2z} G_z} + \underbrace{\chi_{2x} F_x (\sigma_A \cdot \sigma_F)}_{=0} + \underbrace{\chi_{2y} \cdot 0}_{=0} = 4\chi_{2z} G_z \quad \forall j. \quad (28)$$

The key identity is $\sigma_P \cdot \sigma_Q = \sum_k \sigma_P[k] \sigma_Q[k]$: a dot product that collapses the k -sum to a single number, destroying all site-index structure. No chi parameterisation can circumvent this because it is a consequence of the topology (completeness) of the bond graph, not of the coupling strengths.

What would be required is a *site-dependent* effective field with σ_C pattern — for instance, a bond tensor that couples Fe_k only to the Tm sites in its own orbit with an orbit-dependent sign. No such term is present in the Pbnm-symmetry-allowed bilinear chi Hamiltonian.

Option C: Higher-order (χ^2) pathways

Nonlinear double-chi pathways can produce σ_C from σ_F by quadratic combination $(\sigma_C \circ \sigma_F) \circ \sigma_C = \sigma_C$ (since $\sigma_C \circ \sigma_F = 0$... this does not help). A useful combination requires, e.g., $(\sigma_G \circ \sigma_G) \circ (\sigma_A \circ \sigma_A) = \sigma_F \circ \sigma_F = \sigma_F$ — still σ_F . In fact, any even power of σ_G is σ_F , so nonlinear pathways through chi2z alone also cannot produce σ_C pattern by symmetry.

Option D: $H \parallel a$ or $H \parallel b$ geometry

Under $H \parallel a$ the detector couples to $M_x^{\text{Tm}} = c_2(\sigma_A \cdot \lambda_{\text{vec}}^2)$. With chi2z generating $\lambda^2 \sim \sigma_F$:

$$M_x^{\text{Tm}} = c_2 \bar{\lambda}^2 (\sigma_A \cdot \sigma_F) = c_2 \bar{\lambda}^2 \cdot 0 = 0. \quad (29)$$

Likewise $M_y^{\text{Tm}} = c_2(\sigma_G \cdot \sigma_F) = 0$.

All three detection geometries fail for chi2z in Γ_2 . The cross-peak (ω_{qAFM} , $\omega_{E_{12}}$) is forbidden regardless of probe polarisation as long as χ_{2z} is the only active coupling and the Fe order is G -type.

8 Numerical verification

The analytical predictions were verified numerically by the diagnostic executable `diag_tmfeo3_qafm_e12_xpeak` (built from `src/apps/`, target `diag_tmfeo3_qafm_e12_xpeak`).

Ground state ($\chi_{2z} = 0.6$, **Fe pinned to Γ_2**). After 2000 deterministic exact-diag sweeps:

Tm site	$H[\lambda^2]$	λ^2	$\eta_{\text{Tm}}^z \cdot \lambda^2$
0	+11.9999	−0.9967	−0.9967
1	+11.9999	−0.9967	+0.9967
2	+11.9999	−0.9967	+0.9967
3	+11.9999	−0.9967	−0.9967
σ_F -sum	+47.9996	−3.9869	—
σ_C -sum ($= M_z^{\text{Tm}}$)	—	—	0.0000

The $H[\lambda^2]$ values are identical at all four sites, confirming σ_F -pattern. $M_z^{\text{Tm}} = 0.0000$ to numerical precision.

$\chi_{2x} = 0.6$, **same Fe background**. All $H[\lambda^2]_j \approx -3 \times 10^{-6}$ (numerical noise). $\lambda^2 \approx 10^{-5}$ at all sites — chi2x is effectively dead.

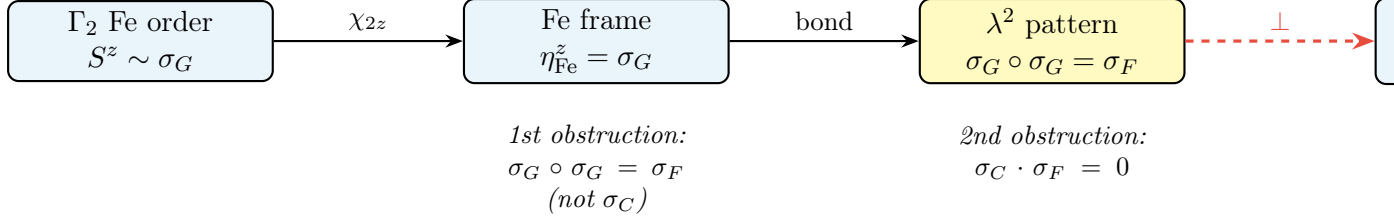
qAFM kick dynamics ($\delta\theta = 0.04\text{rad}$, **chi2z**). Euler integration, $dt = 0.003\text{ps}$, $N = 200$ steps:

- $G_x(t)$ oscillates between ± 0.19 (qAFM is excited).
- $\bar{\lambda}^2(t) = -0.9967$ throughout (chi2z shifts the equilibrium but the amplitude does not change with G_x , only with G_z).
- $M_z^{\text{Tm}}(t) = 0.0000$ at *every* time step, to all displayed digits.

This directly demonstrates that the cross-peak has zero amplitude.

9 Summary diagram

The full selection-rule chain for the $(\omega_{\text{qAFM}}, \omega_{E_{12}})$ cross-peak under $H \parallel c$ in TmFeO_3 Γ_2 can be summarised as a “double-orthogonality trap”:



Constraint	Expression
Fe frame produces σ_F lambda	$\eta_{\text{Fe}}^z \circ S_{\text{Fe}}^z\text{-pattern} = \sigma_G \circ \sigma_G = \sigma_F$
Detector is blind to σ_F	$\eta_{\text{Tm}}^z \cdot \sigma_F = \sigma_C \cdot \sigma_F = 0$
No geometry rescues it	$\sigma_A \cdot \sigma_F = \sigma_G \cdot \sigma_F = \sigma_C \cdot \sigma_F = 0$
Complete bond graph locks $\lambda^2 \rightarrow \sigma_F$	$\sum_k \sigma_P[k] \sigma_Q[k]$ is j -independent
No other chi channel helps	See Table 1

A Bond-by-bond verification: each Tm sees all four Fe

From `fe_tm_w_bond_pairs()` (even bonds only), each Tm sublattice is connected to all four Fe sublattices, one per orbit. The product $\eta_{\text{Fe},k}^z \cdot \sigma_G[k]$ for each bond equals +1 regardless of k (since $\sigma_G[k]^2 = 1$), confirming the uniform field $H[\lambda^2]_j = 4\chi_{2z}G_z$ for all j .

Tm site	Orbit	Fe bonded	$\eta_{\text{Fe},k}^z$	$\sigma_G[k]$	$\eta^z \cdot \sigma_G$	Contribution
Tm ₀	1	Fe ₃	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	2	Fe ₂	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	3	Fe ₁	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	4	Fe ₀	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	Total $H[\lambda^2]_0$					$4\chi_{2z}G_z$
Tm ₁	1	Fe ₂	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	2	Fe ₃	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	3	Fe ₀	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	4	Fe ₁	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	Total $H[\lambda^2]_1$					$4\chi_{2z}G_z$
Tm ₂	1	Fe ₁	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	2	Fe ₀	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	3	Fe ₃	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	4	Fe ₂	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	Total $H[\lambda^2]_2$					$4\chi_{2z}G_z$
Tm ₃	1	Fe ₀	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	2	Fe ₁	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	3	Fe ₂	+1	+1	+1	$+1 \cdot \chi_{2z}G_z$
	4	Fe ₃	−1	−1	+1	$+1 \cdot \chi_{2z}G_z$
	Total $H[\lambda^2]_3$					$4\chi_{2z}G_z$

The pattern across $(j = 0, 1, 2, 3)$ is $(4, 4, 4, 4)\chi_{2z}G_z = \sigma_F$ (uniform), confirming Eq. (9).

B Sigma vector inner product table

$\sigma_P \cdot \sigma_Q$	σ_F	σ_G	σ_A	σ_C	$\sigma_P \circ \sigma_Q$	σ_F	σ_G	σ_A	σ_C
σ_F	4	0	0	0	σ_F	σ_F	σ_G	σ_A	σ_C
σ_G	0	4	0	0	σ_G	σ_G	σ_F	σ_C	σ_A
σ_A	0	0	4	0	σ_A	σ_A	σ_C	σ_F	σ_G
σ_C	0	0	0	4	σ_C	σ_C	σ_A	σ_G	σ_F

The Hadamard product table (right) forms a copy of the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$, with σ_F as the identity element.